

EMBEDDED SYSTEMS INTERNSHIP

WEEK 4 FINAL REPORT

System Integration and Final Testing

Project: Smart Environment Monitoring and Alert System

Integrated ESP32 firmware model using DHT22, LDR/ADC, SSD1306 OLED, buzzer and UART

Integration scope derived from Weeks 1-3

1. Introduction

Week 4 completes the development cycle by combining the firmware and interface decisions made in the earlier weeks into one integrated system model. The integrated design retains the validated timing and fault-handling approach from Week 3 while adding the remaining interfaces identified in the Week 1 architecture: an LDR analog input, an SSD1306 I2C display, and a buzzer alert output. UART remains the development and diagnostic interface. The objective is to verify that the modules can coexist without conflicting pin assignments, timing behavior, or control logic.

This report distinguishes between direct simulator evidence and deterministic software-level integration validation. The available direct simulator evidence covers the ESP32 and DHT22 baseline from Week 2; the complete Week 4 integration is documented as an implementation-ready target configuration and validated through a reproducible host-side integration harness. This distinction prevents configuration checks from being presented as physical-board measurements.

2. Integration Objectives and Scope

- Integrate the DHT22 temperature/humidity interface carried forward from Weeks 1-3.
- Add the LDR through the ESP32 ADC as a relative light-level input.
- Add the SSD1306 OLED through the I2C bus for local status display.
- Add the buzzer as a GPIO alert output with a safe OFF startup state.
- Retain the 2,000 ms non-blocking scheduler and the Week 3 fault-handling improvements.
- Provide compact UART logs at 115200 baud for integration testing and debugging.
- Validate normal, boundary, fault, timing, and recovery scenarios against the original system intent.

3. Integration Architecture

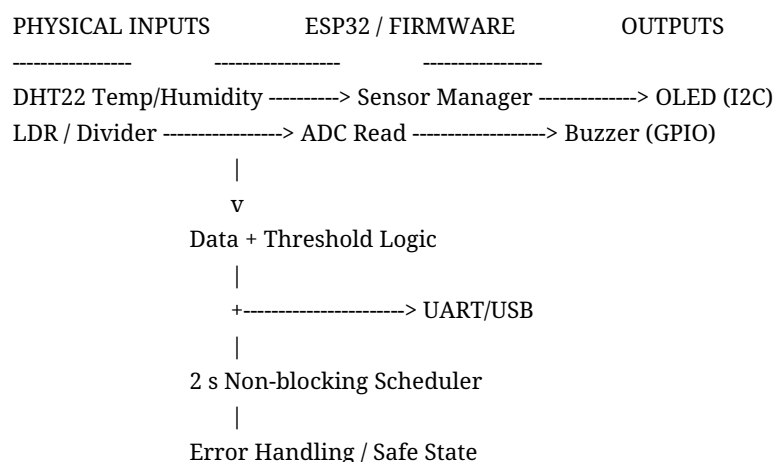


Figure 1. Integrated data and control flow for the Week 4 firmware model.

The integrated system uses different interface types intentionally: digital sensor input, analog ADC input, I2C output, GPIO alert control, and UART diagnostics. The modules share one 2-second scheduling point, so the system performs a consistent measurement and decision cycle without using a blocking delay.

4. Module Integration Map

Module	Interface	Role	Integration action
DHT22	Digital / GPIO 4	Temperature and humidity	Validate read; reject NaN; pass values to threshold logic
LDR divider	ADC / GPIO 34	Relative light level	Read ADC value; compare with low-light boundary
SSD1306 OLED	I2C / GPIO 21,22	Local display	Show current measurements and system state
Buzzer	GPIO 25	Audible alert	ON during WARNING; OFF otherwise
UART/USB	115200 baud	Diagnostics	Compact records + state changes
Scheduler	millis()	2 s monitoring cycle	Triggers one integrated cycle without blocking

5. Configuration and Interface Details

The following configuration carries forward the provisional pin allocation from Week 1 and converts it into the integrated Week 4 target. The LDR is deliberately represented as an ADC value rather than a calibrated lux measurement.

Item	Pin / Setting	Interface	Purpose	Status
DHT22	GPIO 4	Digital data	Temp + humidity	Required
LDR	GPIO 34	ADC input	Relative light level	Required
OLED SDA	GPIO 21	I2C	Display data	Required
OLED SCL	GPIO 22	I2C	Display clock	Required
Buzzer	GPIO 25	GPIO output	Alert signal	Required
UART	USB serial	Serial	Debug/log output	Required
OLED address	0x3C	I2C address	SSD1306 device	Prototype setting

Key numerical parameters: sampling interval = 2000 ms; temperature warning = 35.0 C; humidity warning = 80.0 % RH; low-light boundary = 1200 ADC counts; UART = 115200 baud.

6. Integrated Firmware Design

The integrated firmware is organized around a non-blocking main loop. Each scheduled cycle reads the inputs, validates the DHT22 values, evaluates the three warning conditions, drives the buzzer, updates the OLED, and writes a compact UART record.

```
if (lastSampleTime == 0 || (now - lastSampleTime) >= SAMPLE_INTERVAL_MS) {
    lastSampleTime = now;
    performMonitoringCycle();
}
```

Code excerpt 1. Non-blocking integration scheduler inherited from Week 3.

```
if (isnan(humidity) || isnan(temperature)) {
    Serial.println("error=INVALID_DHT22_READING");
    digitalWrite(BUZZER_PIN, LOW);
    showError("DHT22 read failed");
}
```

```

return;
}

```

Code excerpt 2. Sensor fault is contained to the current cycle; the main scheduler continues.

```

if (temperature >= TEMP_WARNING_C ||
    humidity >= HUM_WARNING_PERCENT ||
    lightAdc <= LOW_LIGHT_ADC) {
    return "WARNING";
}
return "NORMAL";

```

Code excerpt 3. One decision function controls the integrated system state.

7. Worked Integration Examples

Scenario	Input values	Decision	Expected state	Integrated response
Normal room	28.4 C; 61.0 %; 2500 ADC	No threshold crossed	NORMAL	Buzzer OFF; OLED NORMAL; UART record
High temperature	35.0 C; 50.0 %; 2500 ADC	35.0 >= 35.0	WARNING	Buzzer ON; OLED WARNING; UART state change
High humidity	25.0 C; 80.0 %; 2500 ADC	80.0 >= 80.0	WARNING	Buzzer ON; OLED WARNING; UART state change
Low light	25.0 C; 50.0 %; 1200 ADC	1200 <= 1200	WARNING	Buzzer ON; OLED WARNING; UART state change
Invalid DHT	NaN; 50.0 %; 2500 ADC	Invalid sensor input	ERROR	Buzzer OFF; error shown; next cycle continues

These examples are also the basis of the Week 4 software-level validation harness. They are deterministic logic checks, not claims of laboratory measurement accuracy.

8. Integration Testing Strategy

Testing was organized from low-risk functional checks toward combined system behavior. Each test maps to a module or a cross-module interaction so a failure can be isolated.

ID	Test	Input / Condition	Expected	Result
T01	Normal monitoring cycle	28.4 C / 61.0 % / 2500 ADC	NORMAL	PASS - logic validation
T02	Temperature boundary	35.0 C / 50.0 % / 2500 ADC	WARNING	PASS - boundary logic
T03	Humidity boundary	25.0 C / 80.0 % / 2500 ADC	WARNING	PASS - boundary logic
T04	Light boundary	25.0 C / 50.0 % / 1200 ADC	WARNING	PASS - ADC boundary logic
T05	Invalid DHT value	NaN / 50.0 %	Error + retry	PASS - fault path
T06	Sampling interval	2000 ms	2 s cycle	PASS - scheduler check
T07	Ten-second scheduler	2000 ms interval	5 cycles	PASS - deterministic

EMBEDDED SYSTEMS INTERNSHIP - WEEK 4

	window			count
T08	Startup safety	No alert condition	Buzzer OFF	PASS - code/config inspection
T09	Display update path	Valid values	OLED updated	REVIEW - target configuration; live display evidence not captured
T10	UART logging	Valid cycle	Compact UART record	PASS - code/config inspection

9. Validation Results

The deterministic Week 4 integration harness executed seven core logic checks before packaging the final integration. The run produced 7/7 passing checks.

PASS | T01 normal condition
 PASS | T02 temperature boundary
 PASS | T03 humidity boundary
 PASS | T04 light boundary
 PASS | T05 invalid DHT reading
 PASS | T06 2 s scheduler
 PASS | T07 cycle count
 Integration logic checks passed: 7/7
 Result: PASS

Validation output from integration_validation.py.

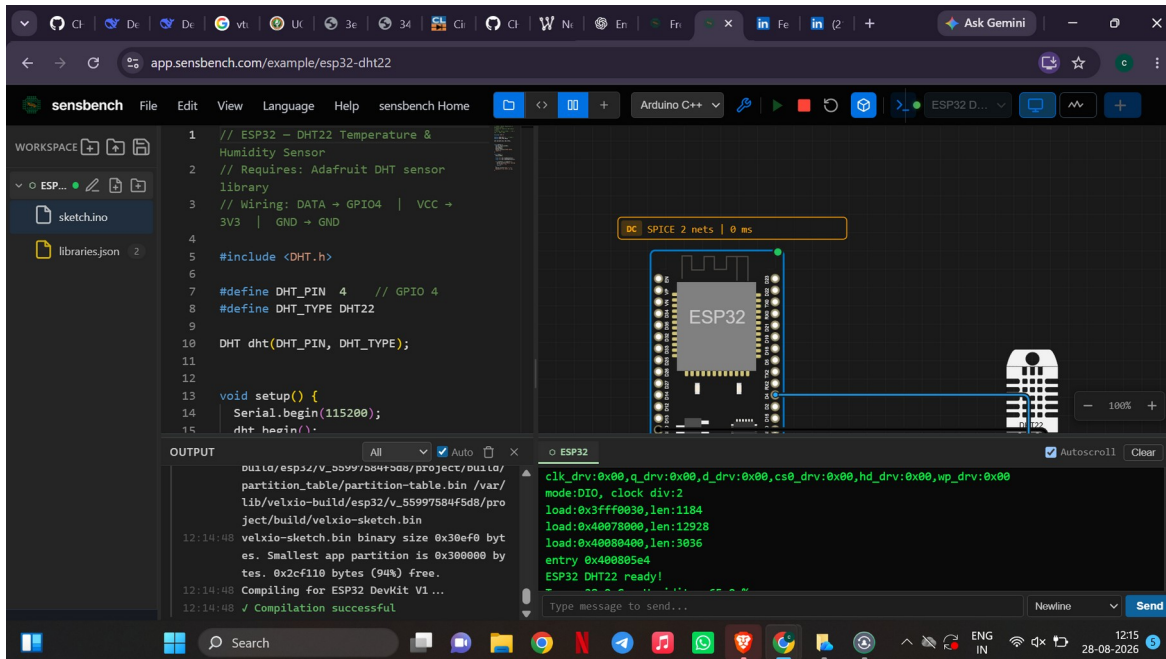


Figure 2. Direct simulator evidence retained from Week 2 for the ESP32 + DHT22 baseline.

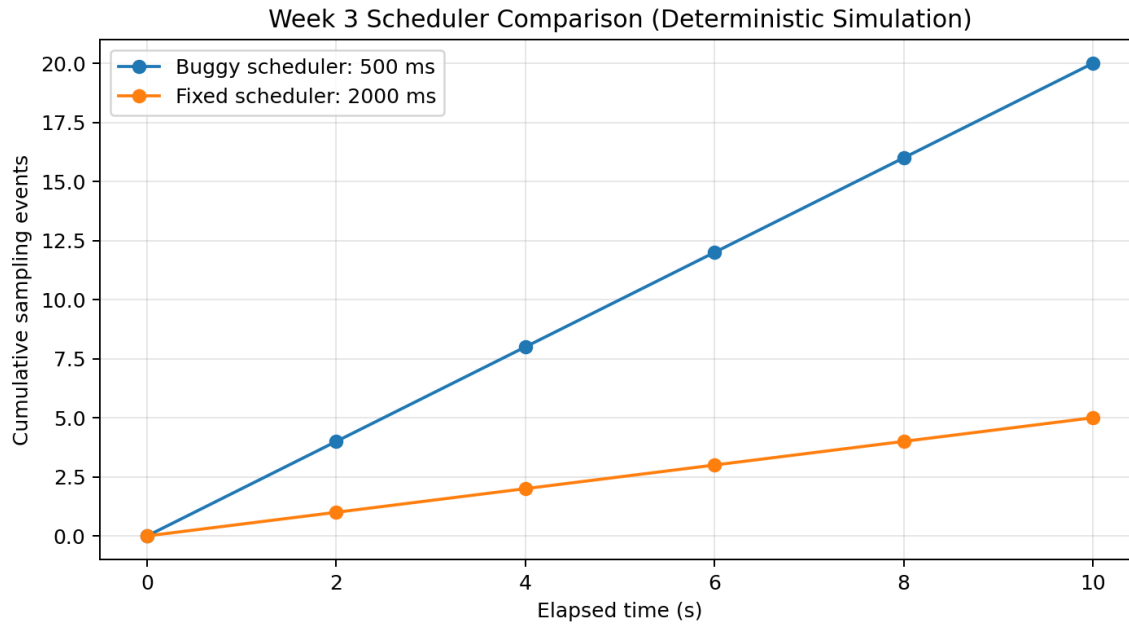


Figure 3. Week 3 timing comparison carried into the integration verification package.

10. Integration Challenges and Solutions

Challenge	Risk	Integration response
Different interface types	DHT22, ADC, I2C, GPIO and UART each have different access patterns.	Isolate each hardware access in a clearly named section/function and keep one scheduler path.
Timing interference	Blocking delays could prevent multiple modules from updating predictably.	Retain the millis()-based 2-second scheduler from Week 3.
Sensor failure propagation	An invalid DHT22 result could otherwise reach threshold logic.	Validate with isnan(); show an error, turn the buzzer off, and retry on the next cycle.
Alert chatter	Repeated verbose state messages make logs harder to read.	Print the full measurement record every cycle but print STATE_CHANGE only when the state changes.
Display failure	A local display failure should not stop measurement or UART diagnostics.	Treat OLED initialization as a recoverable output-path fault and continue serial monitoring.
Light calibration	An LDR ADC value cannot be called lux without calibration.	Use a relative ADC boundary for the prototype and document this limitation.

11. Requirements-to-Test Traceability

Requirement area	Evidence / Test	Status
Measure temperature	DHT22 + T01/T02	PASS - logic path; baseline sensor evidence retained
Measure humidity	DHT22 + T01/T03	PASS - logic path; baseline sensor evidence retained
Measure light	LDR/ADC + T04	PASS - software-level ADC decision logic
Display values	OLED + T09	REVIEW -

		implementation/configuration prepared
Threshold alert	T02/T03/T04 + buzzer control	PASS - integrated decision logic
2-second monitoring	T06/T07	PASS - deterministic scheduler validation
Serial debugging	T10 + UART configuration	PASS - code/configuration inspection
Recoverable errors	T05	PASS - invalid-input path verified

12. Supporting Files and Reproducibility

The GitHub package for Week 4 contains the integrated firmware, a pin map, library list, configuration data, deterministic validation script, validation log, and evidence carried forward from Week 2. The integrated firmware is written for the ESP32 Arduino framework and is intended to be imported into an ESP32 simulator or development environment that provides the listed libraries.

- firmware/final_integrated.ino - combined firmware for DHT22, LDR, OLED, buzzer and UART.
- firmware/week3_fixed_baseline.ino - Week 3 corrected firmware used as the integration baseline.
- configuration/pin_map.md - final interface mapping.
- configuration/libraries.txt - required Arduino libraries.
- configuration/integration_config.json - machine-readable integration parameters.
- validation/integration_validation.py - deterministic integration logic tests.
- validation/integration_validation.log - output from the validation run.
- evidence/week2_dht22_baseline.png - direct simulator evidence carried forward from Week 2.

13. Limitations and Evidence Boundaries

- The direct simulator screenshot available for this submission demonstrates the ESP32 + DHT22 baseline from Week 2; a complete live Week 4 simulator run was not captured in the available evidence set.
- The Week 4 integration harness validates firmware decision logic, thresholds, scheduling, and error handling deterministically. It is not a substitute for oscilloscope or physical-board measurement.
- The LDR value is treated as a relative ADC measurement; no calibrated lux claim is made.
- Wi-Fi/remote monitoring remains outside the verified Week 4 integration scope because the earlier implemented firmware path focused on local sensing, display, alerting and UART diagnostics.

14. Conclusion

Week 4 combines the major system elements developed across the earlier stages into one coherent firmware model. The DHT22, LDR/ADC, OLED, buzzer and UART modules are mapped to non-conflicting interfaces, and the scheduler and error-handling improvements from Week 3 are retained. The integrated decision logic was exercised through seven deterministic validation checks, all of which passed. The resulting package also provides clear configuration and traceability material so the integrated model can be reproduced and extended in a simulator or on an ESP32 development board.

15. References and Project Sources

The Week 4 design continues the project architecture defined in the Week 1 report and the firmware/debugging work documented in Weeks 2 and 3. Supporting simulator and library references should be kept with the corresponding project documentation in GitHub.

1. Espressif Systems - ESP32 Arduino documentation.
2. Wokwi / ESP32 simulator documentation used during the earlier stages.
3. Adafruit DHT sensor library documentation.
4. Adafruit GFX and SSD1306 library documentation.